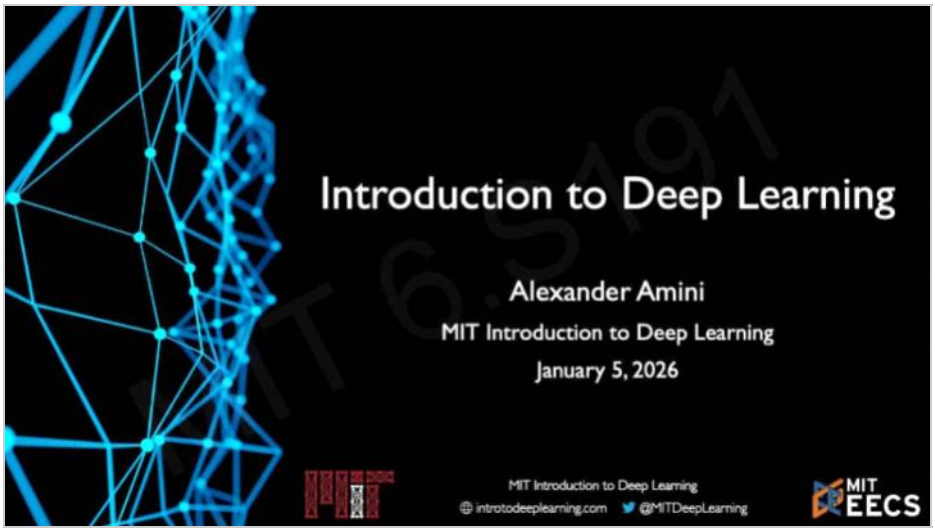
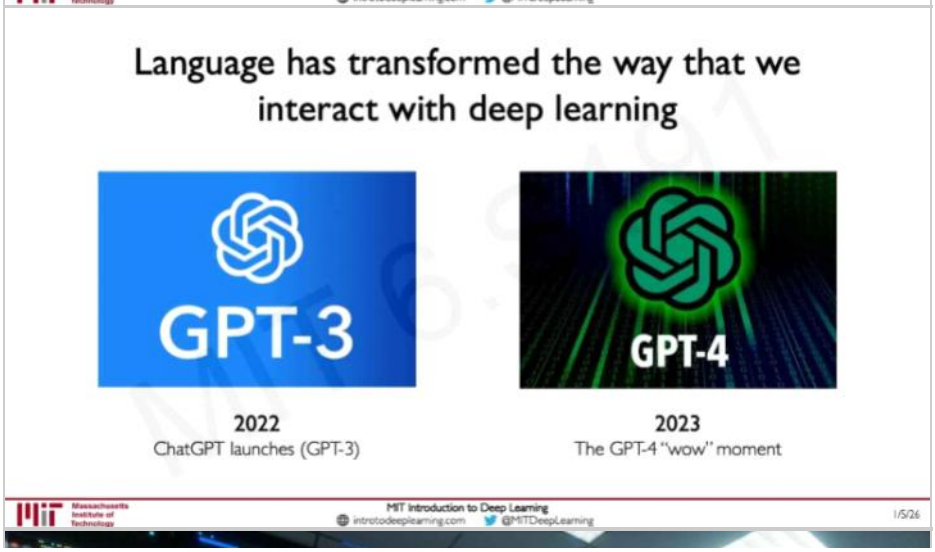
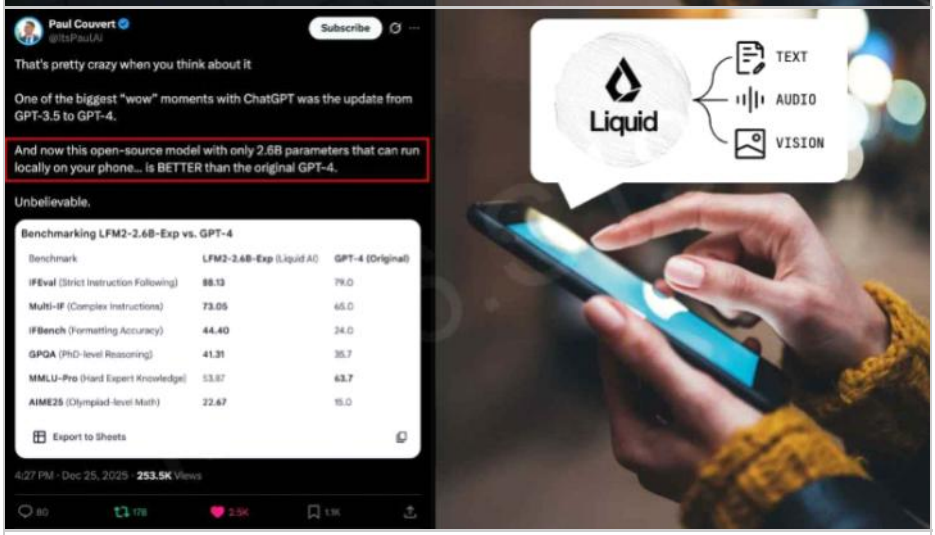
Lecture 1

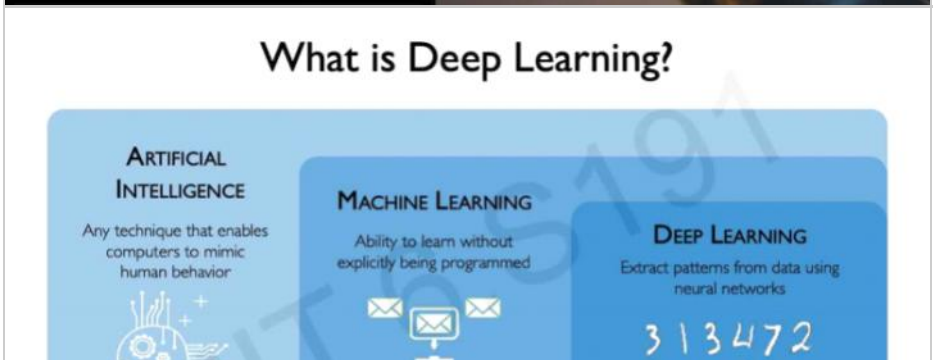
Thursday, 2 April 2026 9:43 am

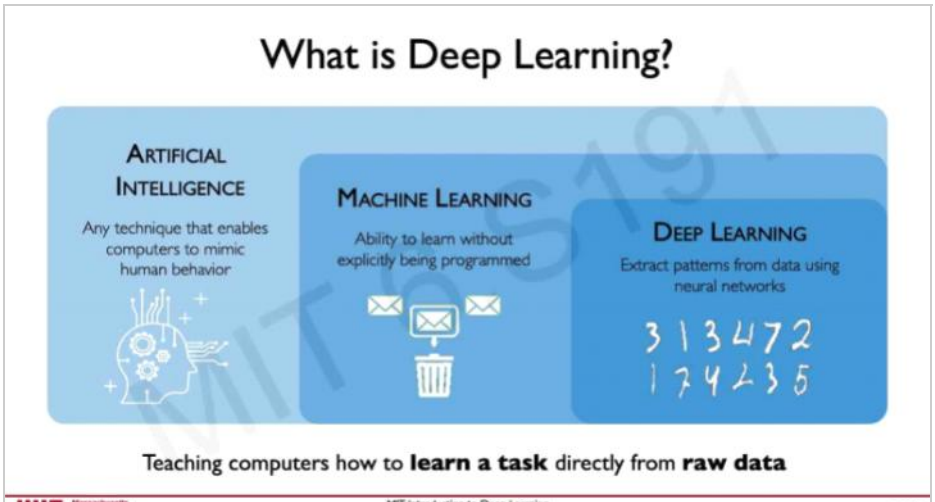
6S191_MIT _DeepLea...






Paul Couvert
@ItsPaulAi
Subscribe
That's pretty crazy when you think about it
One of the biggest "wow" moments with ChatGPT was the update from GPT-3.5 to GPT-4.
And now this open-source model with only 2.6B parameters that can run locally on your phone... is BETTER than the original GPT-4.
Unbelievable.
Benchmarking LFM2-2.6B-Exp vs. GPT-4
Benchmark | LFM2-2.6B-Exp (Liquid AI) | GPT-4 (Original)
IFEval (Strict Instruction Following) | 88.13 | 79.0
Multi-IF (Complex Instructions) | 73.05 | 65.0
IFBench (Formatting Accuracy) | 44.40 | 24.0
GPQA (PhD-level Reasoning) | 41.31 | 35.7
MMLU-Pro (Hard Expert Knowledge) | 53.87 | 63.7
AIME25 (Olympiad-level Math) | 22.67 | 15.0
Export to Sheets
4:27 PM · Dec 25, 2025 · 253.5K Views
Liquid
TEXT
AUDIO
VISION


What is Deep Learning?
Artificial Intelligence
Any technique that enables computers to mimic human behavior
Machine Learning
Ability to learn without explicitly being programmed
Deep Learning
Extract patterns from data using neural networks
3 1 3 4 7 2


What is Deep Learning?
Artificial Intelligence
Any technique that enables computers to mimic human behavior
Machine Learning
Ability to learn without explicitly being programmed
Deep Learning
Extract patterns from data using neural networks
Teaching computers how to learn a task directly from raw data
MIT Introduction to Deep Learning
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Lecture Schedule
Intro to Deep Learning
Deep Sequence Modeling
Deep Learning in Python; Music Generation
Deep Computer Vision
Deep Generative Modeling
Facial Detection Systems
Deep Reinforcement Learning
New Frontiers
Fine-Tune an LLM, You Must!
AI for Science
Secrets to Massively Parallel Training
Final Project
Guest Lecture
Project Presentations
6.S191
Deep Learning
• Jan 5 – Jan 9, 2026
• Lecture Breakdown
• Labs: TensorFlow & PyTorch
• Competitions
• Final Projects + Prizes!
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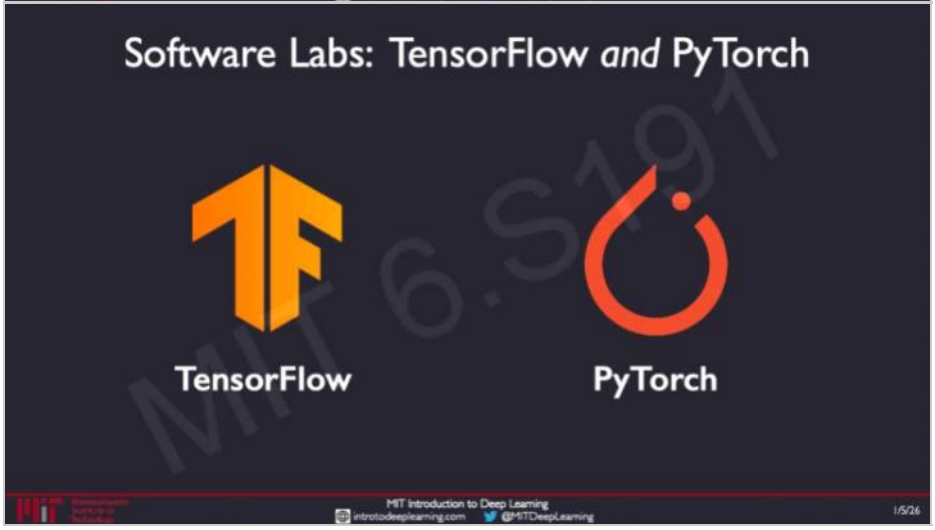
Software Labs: TensorFlow and PyTorch
TensorFlow
PyTorch
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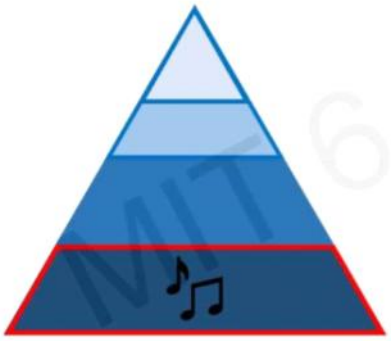
Labs and Prizes

All due Thursday 1/8 at 11:59pm ET. Instructions: bit.ly/6s191-syllabus



Music Generation

Build a neural network that can learn the genre of Irish folk songs and use it to generate brand new songs!

Prize:

Labs and Prizes

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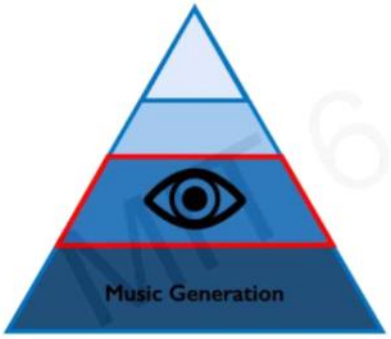

Computer Vision

Build a neural network that can detect and mitigate biases in computer vision facial recognition systems!

Prize:

Labs and Prizes

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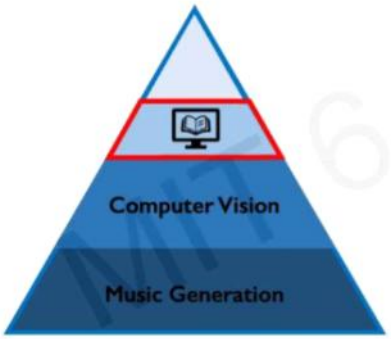


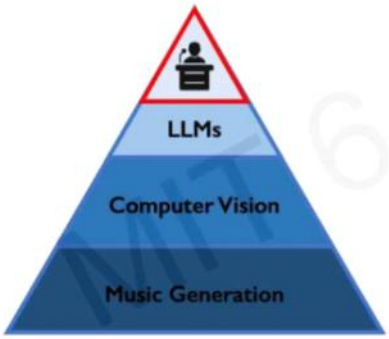
Large Language Models

Finetune the Gemma large language model (LLM) in a mystery style and evaluate with an AI judge!

Prize:

Project Pitch Competition

Friday 1/9. Instructions: bit.ly/6s191-syllabus



Project Pitch Competition

Present a novel deep learning research idea or application (5 minutes, strict)

Presentations on **Friday, Jan 9**

Submit groups by **Wed 1/07 11:59pm ET**

Submit slides by **Thu 1/08 11:59pm ET**

Instructions: bit.ly/6s191-syllabus

Prizes:

Gold: NVIDIA 5070 GPU

Silver: Smartwatch

Bronze: HD Monitor

Program Support

- All lectures will be held in person in 32-123
- Software labs + office hours in 32-123
- Piazza: piazza.com/mit/spring2026/6s191
  - Course announcements & asking questions
- Program Website: introtodeeplearning.com
  - Lecture schedule
  - Slides and lecture recordings
  - Software labs
- Syllabus: bit.ly/6s191-syllabus
- Labs: github.com/MITDeepLearning/introtodeeplearning
- Email us: introtodeeplearning-staff@mit.edu

Program Staff

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Lead TA

Shrika Eddula
Lead TA

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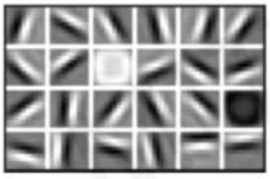
Why Deep Learning and Why Now?

Why Deep Learning?

Hand engineered features are time consuming, brittle, and not scalable in practice

Can we learn the **underlying features** directly from data?

Low Level Features



Lines & Edges

Mid Level Features

Eyes & Nose & Ears

High Level Features

Facial Structure

Why Now?

Neural Networks date back decades, so why the dominance?

1952 Stochastic Gradient Descent

1958 Perceptron
- Learnable Weights

⋮

1986 Backpropagation
- Multi-Layer Perceptron

1995 Deep Convolutional NN
- Digit Recognition

⋮

1. Big Data
- Larger Datasets
- Easier Collection & Storage

IMAGENET

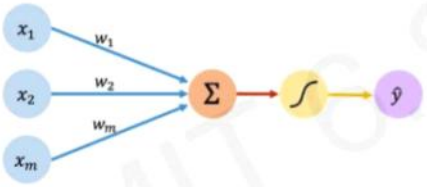
2. Hardware
- Graphics Processing Units (GPUs)
- Massively Parallelizable

3. Software
- Improved Techniques
- New Models
- Toolboxes

The Perceptron

The structural building block of deep learning

The Perceptron: Forward Propagation



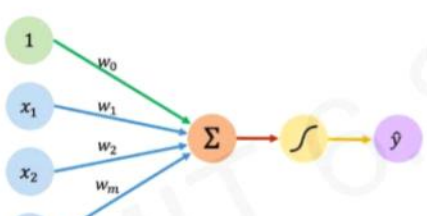
Inputs Weights Sum Non-Linearity Output

Output — Linear combination of inputs

$$\hat{y} = g\left(\sum_{i=1}^{m} x_i\, w_i\right)$$

Non-linear activation function

The Perceptron: Forward Propagation

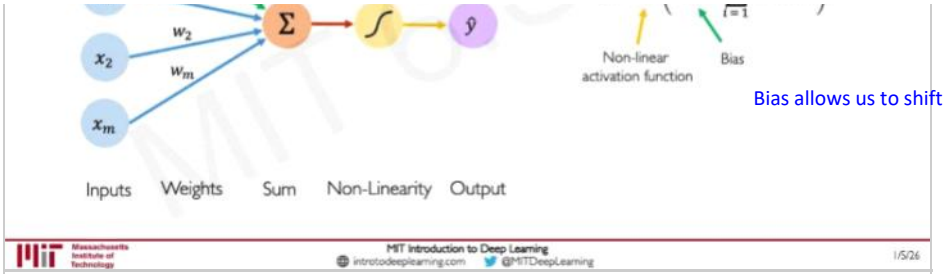


Output — Linear combination of inputs

$$\hat{y} = g\left(w_0 + \sum_{i=1}^{m} x_i\, w_i\right)$$

Non-linear activation function — Bias

Bias allows us to shift function up and down



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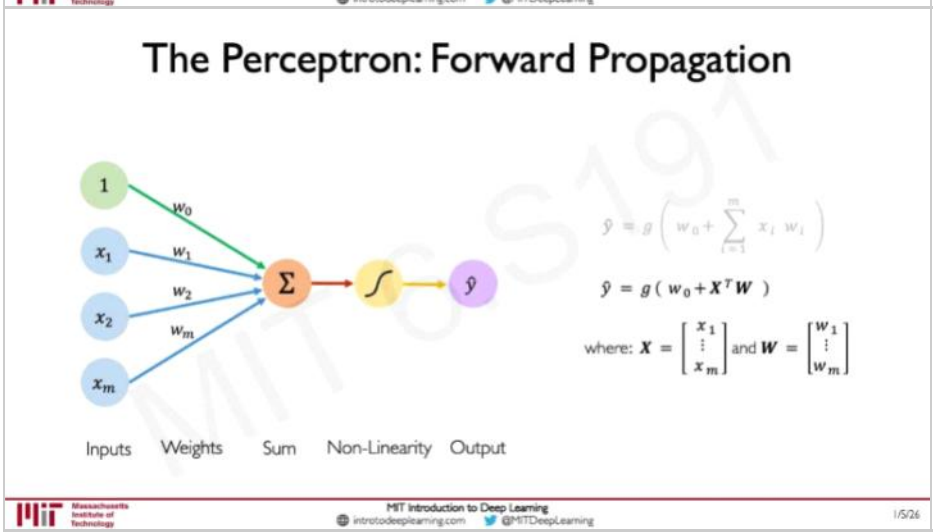
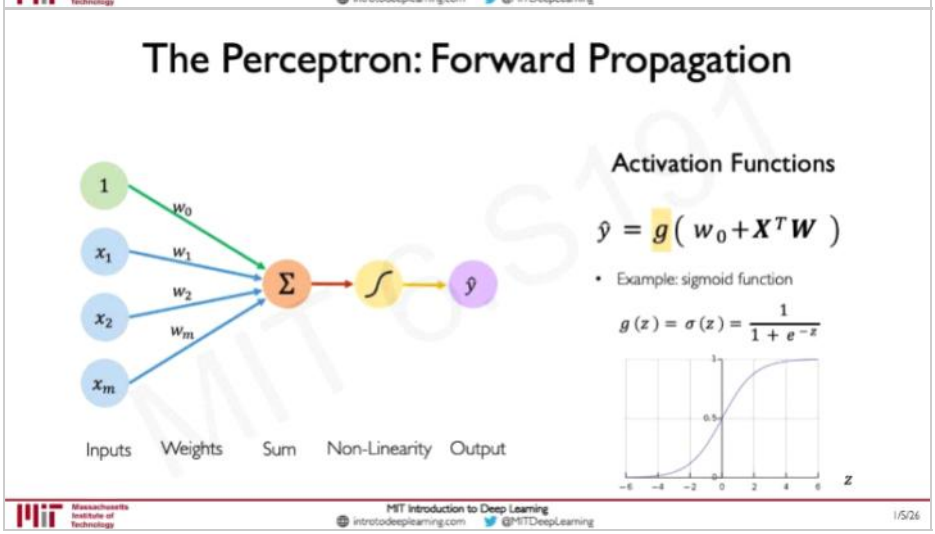


Take dot product between X and W
Add bias
Apply non-linear activation function - Transformation

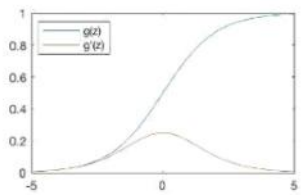


Common Activation Functions

Sigmoid: Mostly used for probabilities (0-1)

ReLU: 0 - inf good for removing negatives

Sigmoid Function



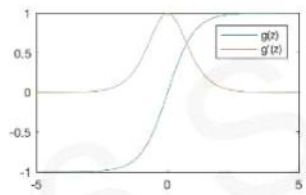
$$g(z) = \frac{1}{1 + e^{-z}}$$

$$g'(z) = g(z)(1 - g(z))$$

```
tf.math.sigmoid(z)
```

```
torch.sigmoid(z)
```

Hyperbolic Tangent



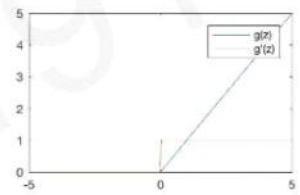
$$g(z) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$$

$$g'(z) = 1 - g(z)^2$$

```
tf.math.tanh(z)
```

```
torch.tanh(z)
```

Rectified Linear Unit (ReLU)



$$g(z) = \max(0, z)$$

$$g'(z) = \begin{cases} 1, & z > 0 \\ 0, & \text{otherwise} \end{cases}$$

```
tf.nn.relu(z)
```

```
torch.nn.ReLU(z)
```

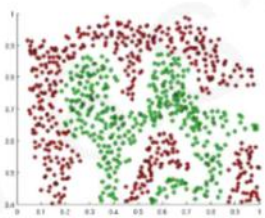
TensorFlow code blocks

NOTE: All activation functions are non-linear

PyTorch code blocks

Importance of Activation Functions

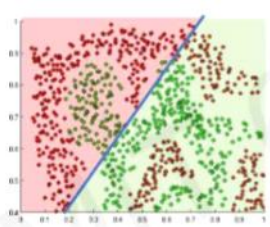
*The purpose of activation functions is to **introduce non-linearities** into the network*



What if we wanted to build a neural network to distinguish green vs red points?

Importance of Activation Functions

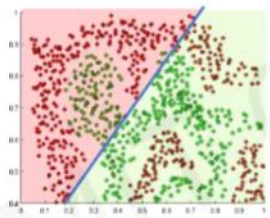
*The purpose of activation functions is to **introduce non-linearities** into the network*



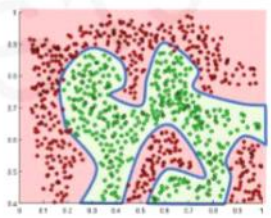
Linear activation functions produce linear decisions no matter the network size

Importance of Activation Functions

The purpose of activation functions is to ***introduce non-linearities*** *into the network*

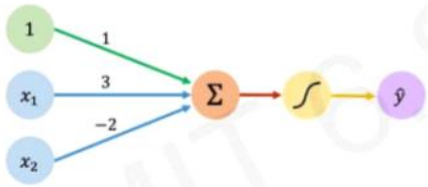


Linear activation functions produce linear decisions no matter the network size



Non-linearities allow us to approximate arbitrarily complex functions

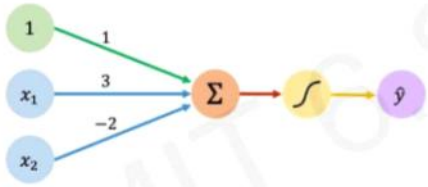
The Perceptron: Example



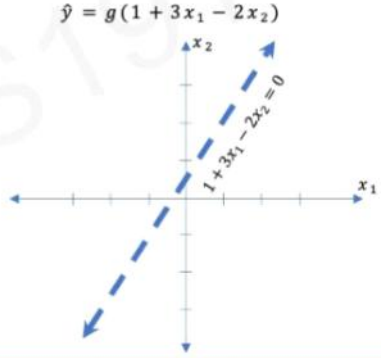
We have: $w_0 = 1$ and $\boldsymbol{W} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$

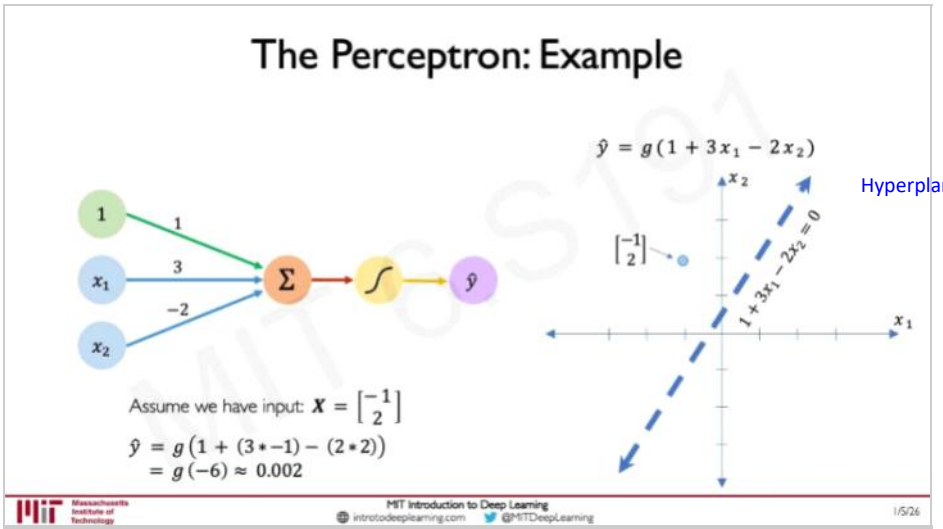
$$\hat{y} = g(w_0 + \boldsymbol{X}^T \boldsymbol{W})$$
$$= g\left(1 + \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^T \begin{bmatrix} 3 \\ -2 \end{bmatrix}\right)$$
$$\hat{y} = g\underbrace{(1 + 3x_1 - 2x_2)}_{\text{This is just a line in 2D!}}$$

The Perceptron: Example

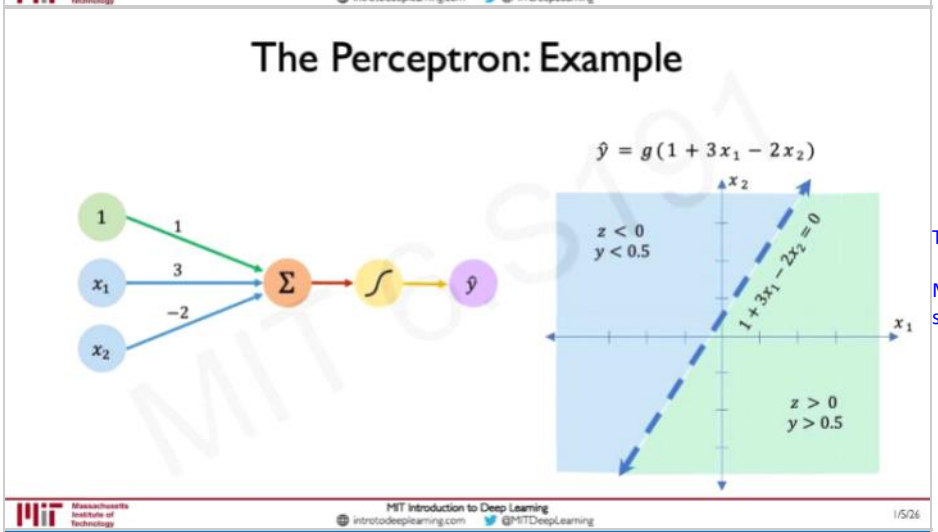


$$\hat{y} = g(1 + 3x_1 - 2x_2)$$



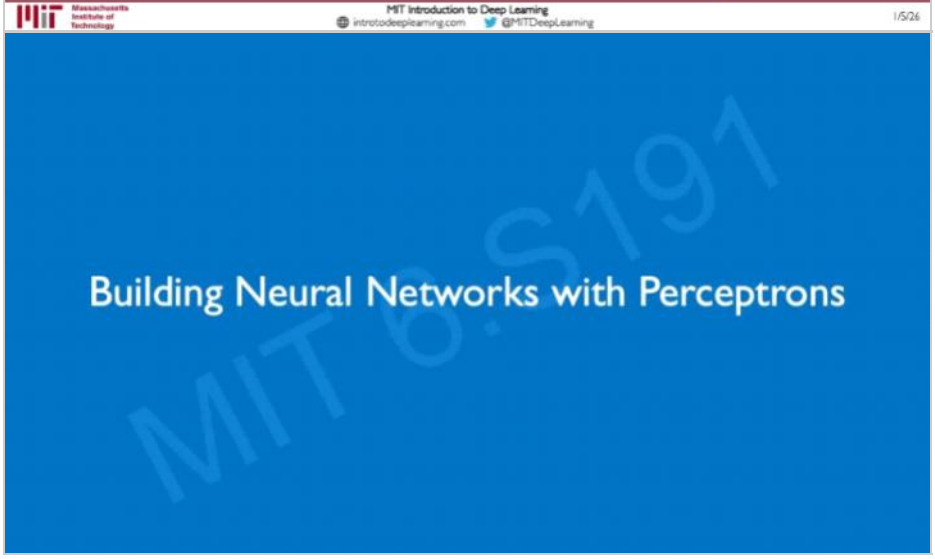


Hyperplane decision boundary



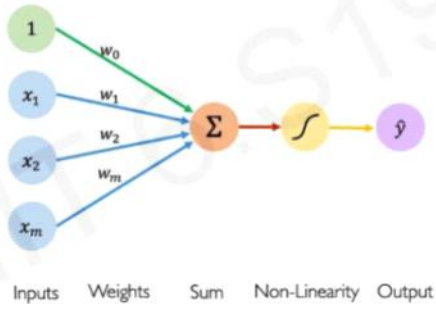
This problem only has 2 parameters.

Modern DL problems like LLMs use 2B parameters…
so imagine a 2B dimensional space - NOT POSSIBLE!



The Perceptron: Simplified

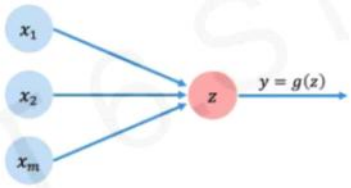
$$\hat{y} = g\left(w_0 + \boldsymbol{X}^T \boldsymbol{W} \right)$$



How a Neuron works - 3 steps:

1. Dot product
2. Add bias
3. Apply activation function

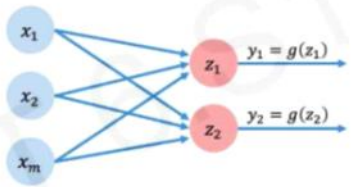
The Perceptron: Simplified



$$z = w_0 + \sum_{j=1}^{m} x_j w_j$$

Multi Output Perceptron

Because all inputs are densely connected to all outputs, these layers are called **Dense** layers



$$z_i = w_{0,i} + \sum_{j=1}^{m} x_j w_{j,i}$$

Dense layer from scratch

```
class MyDenseLayer(tf.keras.layers.Layer):
  def __init__(self, input_dim, output_dim):
    super(MyDenseLayer, self).__init__()

    # Initialize weights and bias
    self.W = self.add_weight([input_dim, output_dim])
    self.b = self.add_weight([1, output_dim])
```

```
class MyDenseLayer(nn.Module):
  def __init__(self, input_dim, output_dim):
    super(MyDenseLayer, self).__init__()

    # Initialize weights and bias
    self.W = nn.Parameter(torch.randn(input_dim,
        output_dim, requires_grad=True)
```

```python
# Initialize weights and bias
self.W = self.add_weight([input_dim, output_dim])
self.b = self.add_weight([1, output_dim])

def call(self, inputs):
  # Forward propagate the inputs
  z = tf.matmul(inputs, self.W) + self.b

  # Feed through a non-linear activation
  output = tf.math.sigmoid(z)
  return output
```

```python
# Initialize weights and bias
self.W = nn.Parameter(torch.randn(input_dim,
    output_dim, requires_grad=True)
self.b = nn.Parameter(torch.randn(1, output_dim,
    requires_grad=True)

def forward(self, inputs):
  # Forward propagate the inputs
  z = torch.matmul(inputs, self.W) + self.b

  # Feed through a non-linear activation
  output = torch.sigmoid(z)
  return output
```


Multi Output Perceptron

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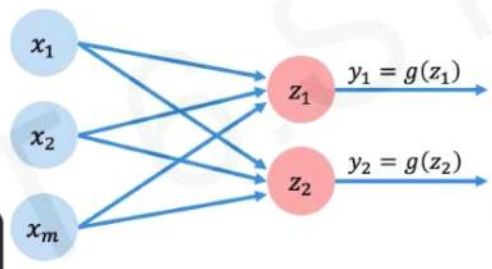
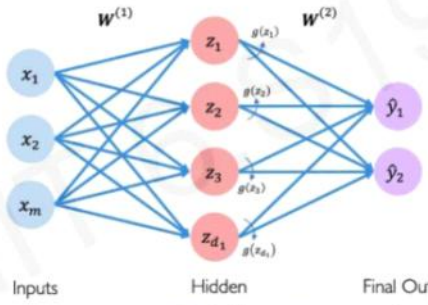



```python
import tensorflow as tf

layer = tf.keras.layers.Dense(
    units=2)
```

```python
import torch.nn as nn

layer = nn.Linear(in_features=m,
    out_features=2)
```

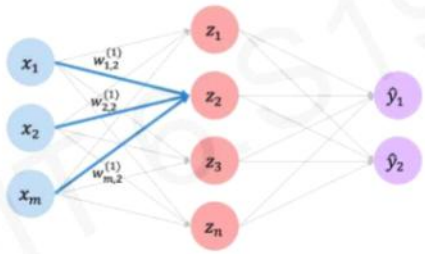
$$z_i = w_{0,i} + \sum_{j=1}^{m} x_j \, w_{j,i}$$

Single Layer Neural Network



$$z_i = w_{0,i}^{(1)} + \sum_{j=1}^{m} x_j \, w_{j,i}^{(1)} \qquad \hat{y}_i = g\left(w_{0,i}^{(2)} + \sum_{j=1}^{d_1} g(z_j)\, w_{j,i}^{(2)}\right)$$

Single Layer Neural Network



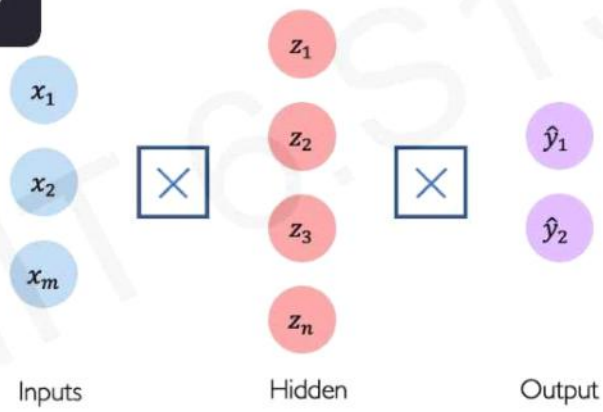
$$z_2 = w_{0,2}^{(1)} + \sum_{j=1}^{m} x_j \, w_{j,2}^{(1)}$$
$$= w_{0,2}^{(1)} + x_1 \, w_{1,2}^{(1)} + x_2 \, w_{2,2}^{(1)} + x_m \, w_{m,2}^{(1)}$$

"deep" has more than 3 layers

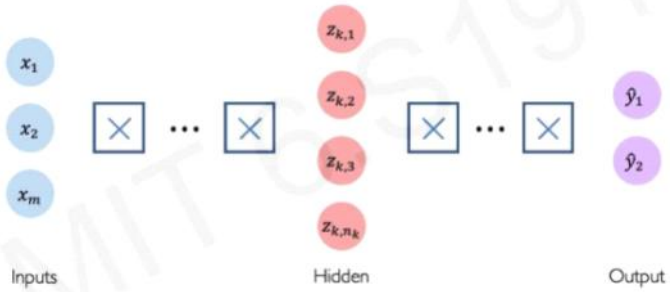
Multi Output Perceptron

```
import tensorflow as tf

model = tf.keras.Sequential([
    tf.keras.layers.Dense(n),
    tf.keras.layers.Dense(2)
])
```

```
from torch import nn

model = nn.Sequential(
    nn.Linear(m, n),
    nn.ReLU(),
    nn.Linear(n, 2)
)
```

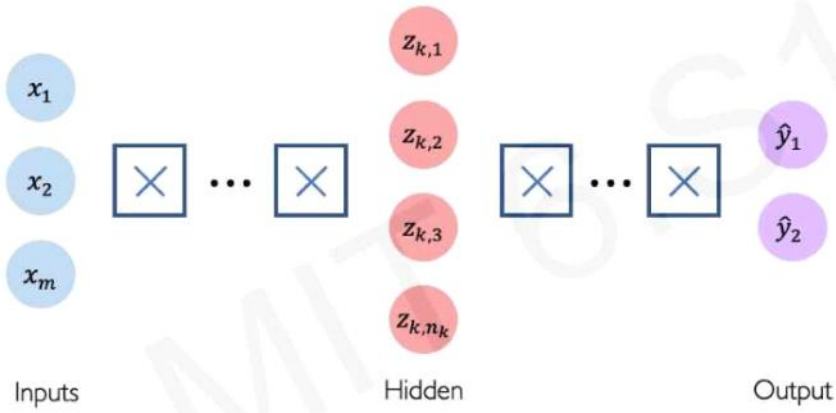


Deep Neural Network



$$z_{k,i} = w_{0,i}^{(k)} + \sum_{j=1}^{n_{k-1}} g(z_{k-1,j}) \, w_{j,i}^{(k)}$$

Deep Neural Network



$$z_{k,i} = w_{0,i}^{(k)} + \sum_{j=1}^{n_{k-1}} g(z_{k-1,j})\, w_{j,i}^{(k)}$$

```python
import tensorflow as tf

model = tf.keras.Sequential([
  tf.keras.layers.Dense(n1),
  tf.keras.layers.Dense(n2),
  ⋮
  tf.keras.layers.Dense(2)
])
```

```python
from torch import nn

model = nn.Sequential(
  nn.Linear(m, n1),
  nn.ReLU(),
⋮
  nn.ReLU(),
  nn.Linear(nK, 2)
)
```


Applying Neural Networks

Example Problem

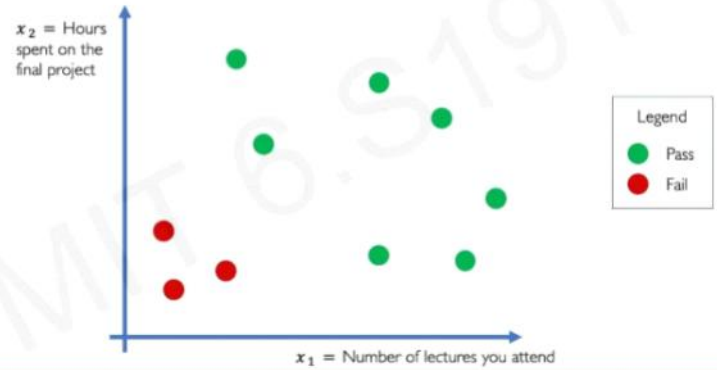
Will I pass this class?

Let's start with a simple two feature model

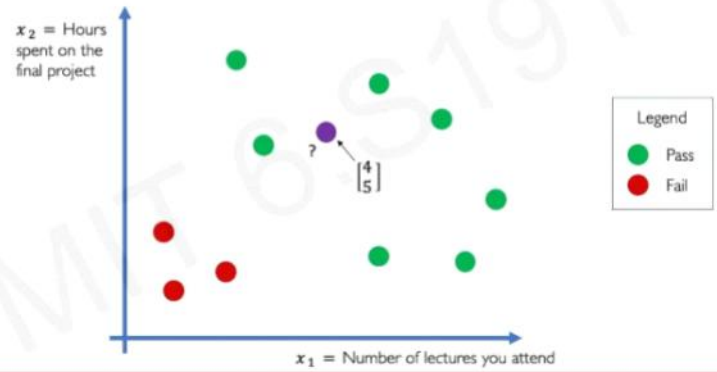
x_1 = Number of lectures you attend

x_2 = Hours spent on the final project

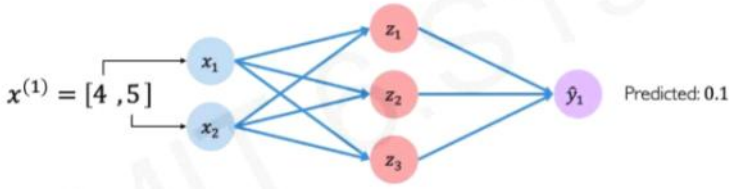
Example Problem: Will I pass this class?



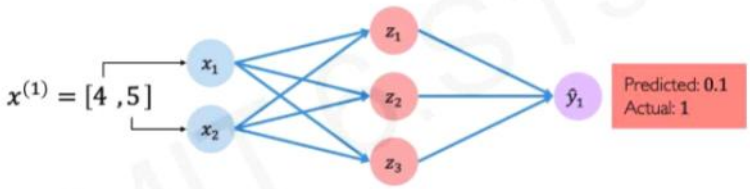
Example Problem: Will I pass this class?



Example Problem: Will I pass this class?

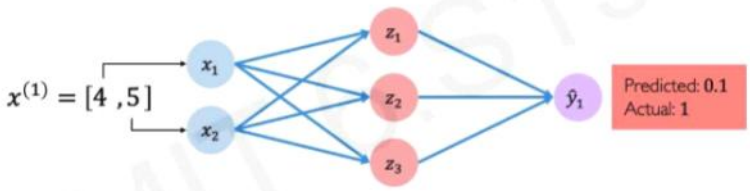


Example Problem: Will I pass this class?



Quantifying Loss

*The **loss** of our network measures the cost incurred from incorrect predictions*

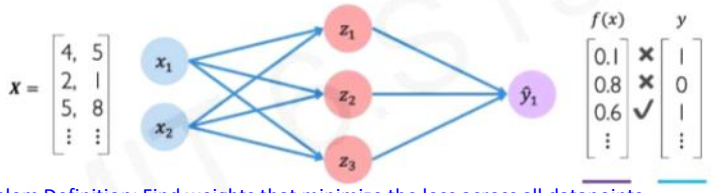


$$\mathcal{L}\big(\underbrace{f\big(x^{(i)}; \boldsymbol{W}\big)}_{\text{Predicted}}, \underbrace{y^{(i)}}_{\text{Actual}}\big)$$

Loss - deviation between predicted and ground truth

Empirical Loss

*The **empirical loss** measures the total loss over our entire dataset*



Problem Definition: Find weights that minimize the loss across all datapoints.

$$J(\boldsymbol{W}) = \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}\big(\underbrace{f\big(x^{(i)}; \boldsymbol{W}\big)}_{\text{Predicted}}, \underbrace{y^{(i)}}_{\text{Actual}}\big)$$

Also known as:
- Objective function
- Cost function
- Empirical Risk

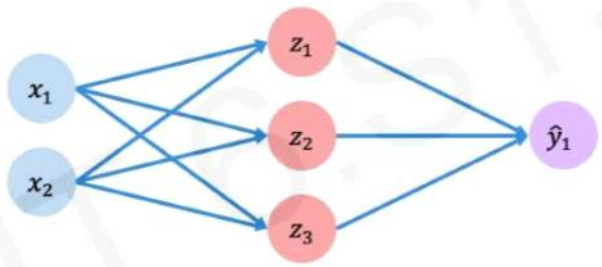
For many datapoints, we compute the average loss over all the predictions

Binary Cross Entropy Loss

For classification problems, where output is binary

Cross entropy loss *can be used with models that output a probability between 0 and 1*

$$X = \begin{bmatrix} 4, & 5 \\ 2, & 1 \\ 5, & 8 \\ \vdots & \vdots \end{bmatrix}$$



$$\begin{array}{cc} f(x) & y \\ \begin{bmatrix} 0.1 \\ 0.8 \\ 0.6 \\ \vdots \end{bmatrix} & \begin{bmatrix} 1 \\ 0 \\ 1 \\ \vdots \end{bmatrix} \end{array}$$

$$J(W) = -\frac{1}{n}\sum_{i=1}^{n} \underbrace{y^{(i)}}_{\text{Actual}} \log\left(\underbrace{f\left(x^{(i)}; W\right)}_{\text{Predicted}}\right) + (1 - \underbrace{y^{(i)}}_{\text{Actual}}) \log\left(1 - \underbrace{f\left(x^{(i)}; W\right)}_{\text{Predicted}}\right)$$

```
loss = tf.reduce_mean( tf.nn.softmax_cross_entropy_with_logits(y, predicted) )
loss = torch.nn.functional.cross_entropy( predicted, y )
```

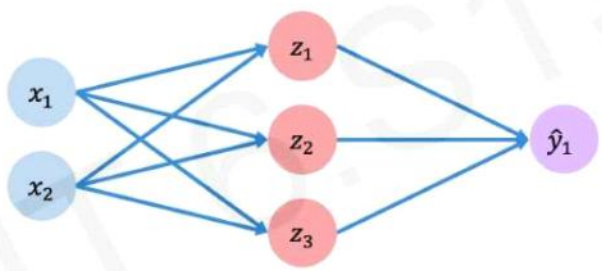
Mean Squared Error Loss

For regressions tasks (with continuous values)

Mean squared error loss *can be used with regression models that output continuous real numbers*

Or L1 error loss

$$X = \begin{bmatrix} 4, & 5 \\ 2, & 1 \\ 5, & 8 \\ \vdots & \vdots \end{bmatrix}$$



$$\begin{array}{cc} f(x) & y \\ \begin{bmatrix} 30 \\ 80 \\ 85 \\ \vdots \end{bmatrix} & \begin{bmatrix} 90 \\ 20 \\ 95 \\ \vdots \end{bmatrix} \end{array}$$

Final Grades (percentage)

$$J(W) = \frac{1}{n}\sum_{i=1}^{n} \left(\underbrace{y^{(i)}}_{\text{Actual}} - \underbrace{f\left(x^{(i)}; W\right)}_{\text{Predicted}}\right)^2$$

```
loss = tf.reduce_mean( tf.square(tf.subtract(y, predicted)) )
loss = tf.keras.losses.MSE( y, predicted )
```

```
loss = torch.nn.functional.mse_loss(
        predicted, y )
```

Training Neural Networks

Loss Optimization

We want to find the network weights that ***achieve the lowest loss***

$$\boldsymbol{W}^* = \underset{\boldsymbol{W}}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}\left(f\left(x^{(i)}; \boldsymbol{W}\right), y^{(i)}\right)$$

$$\boldsymbol{W}^* = \underset{\boldsymbol{W}}{\operatorname{argmin}} J(\boldsymbol{W})$$

Loss is actually a function of the weights

Loss Optimization

We want to find the network weights that ***achieve the lowest loss***

$$\boldsymbol{W}^* = \underset{\boldsymbol{W}}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}\left(f\left(x^{(i)}; \boldsymbol{W}\right), y^{(i)}\right)$$

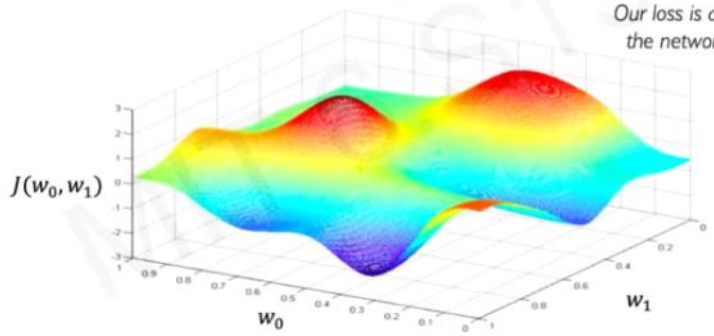
$$\boldsymbol{W}^* = \underset{\boldsymbol{W}}{\operatorname{argmin}} J(\boldsymbol{W})$$

Remember:
$\boldsymbol{W} = \{\boldsymbol{W}^{(0)}, \boldsymbol{W}^{(1)}, \dots\}$

Loss Optimization

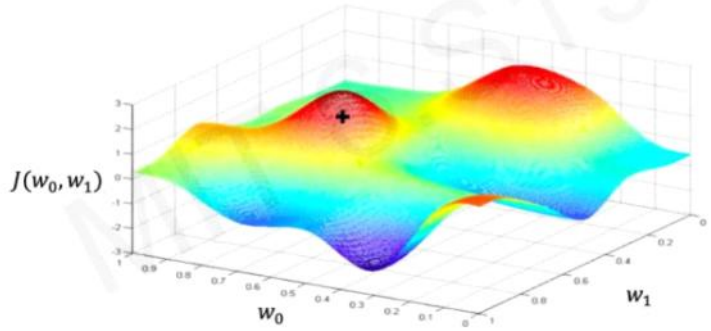
$$W^* = \underset{W}{\operatorname{argmin}}\, J(W)$$

Remember:
Our loss is a function of the network weights!



Loss Optimization

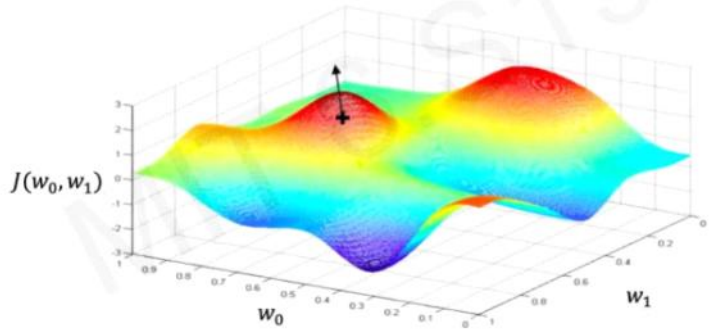
Randomly pick an initial (w_0, w_1)



Loss Optimization

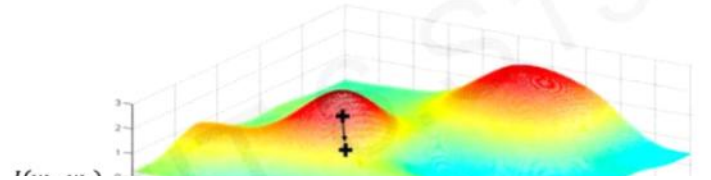
Compute gradient, $\frac{\partial J(W)}{\partial W}$

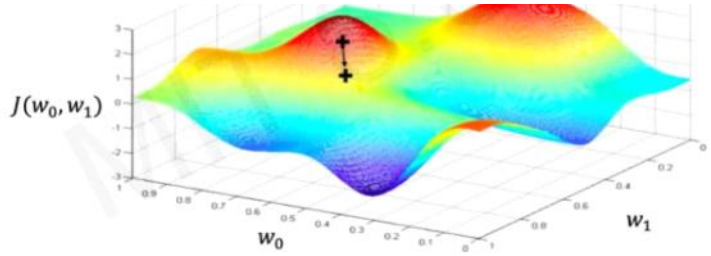
Gradient tells us the direction of the slope at a particular point / instant



Loss Optimization

Take small step in opposite direction of gradient

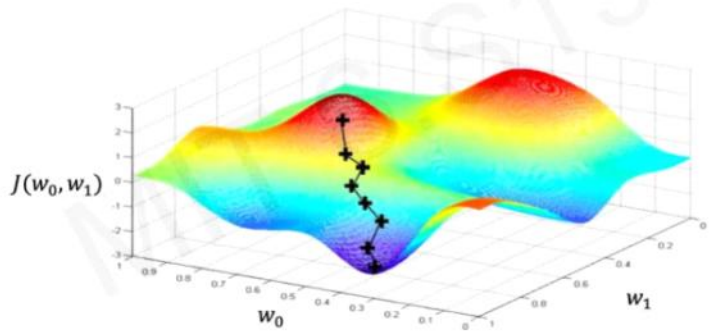




We take small steps in the negative direction (negative slope) to get to the lowest point.

Gradient Descent

Repeat until convergence



Gradient Descent

Algorithm

1. Initialize weights randomly $\sim\mathcal{N}(0, \sigma^2)$
2. Loop until convergence:
3. Compute gradient, $\frac{\partial J(\boldsymbol{W})}{\partial \boldsymbol{W}}$
4. Update weights, $\boldsymbol{W} \leftarrow \boldsymbol{W} - \eta \frac{\partial J(\boldsymbol{W})}{\partial \boldsymbol{W}}$
5. Return weights

Gradient Descent

Algorithm

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3. Compute gradient, $\frac{\partial J(W)}{\partial W}$
4. Update weights, $W \leftarrow W - \eta \frac{\partial J(W)}{\partial W}$
5. Return weights

```python
import tensorflow as tf

weights = tf.Variable([tf.random.normal()])

while True:   # loop forever
  with tf.GradientTape() as g:
    loss = compute_loss(weights)
    gradient = g.gradient(loss, weights)

  weights = weights - lr * gradient
```


Gradient Descent

Algorithm

1. Initialize weights randomly $\sim\mathcal{N}(0, \sigma^2)$
2. Loop until convergence:
3. Compute gradient, $\frac{\partial J(W)}{\partial W}$
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  weights = weights - lr * gradient
```

Massachusetts Institute of Technology
MIT Introduction to Deep Learning
introtodeeplearning.com @MITDeepLearning
1/5/26

Computing Gradients: Backpropagation

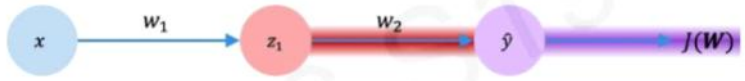
How does a small change in one weight (ex. w_2) affect the final loss $J(W)$?

Computing Gradients: Backpropagation

$$\frac{\partial J(W)}{\partial w_2} =$$

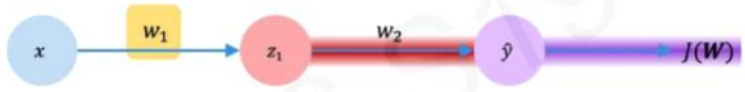
Let's use the chain rule!

Computing Gradients: Backpropagation



$$\frac{\partial J(W)}{\partial w_2} = \frac{\partial J(W)}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial w_2}$$

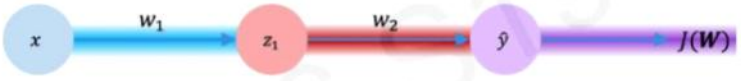
Computing Gradients: Backpropagation



$$\frac{\partial J(W)}{\partial w_1} = \frac{\partial J(W)}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial w_1}$$

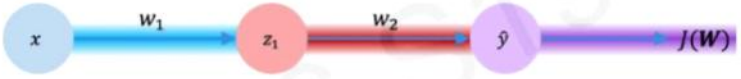
Apply chain rule! Apply chain rule!

Computing Gradients: Backpropagation



$$\frac{\partial J(W)}{\partial w_1} = \frac{\partial J(W)}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial z_1} * \frac{\partial z_1}{\partial w_1}$$

Computing Gradients: Backpropagation



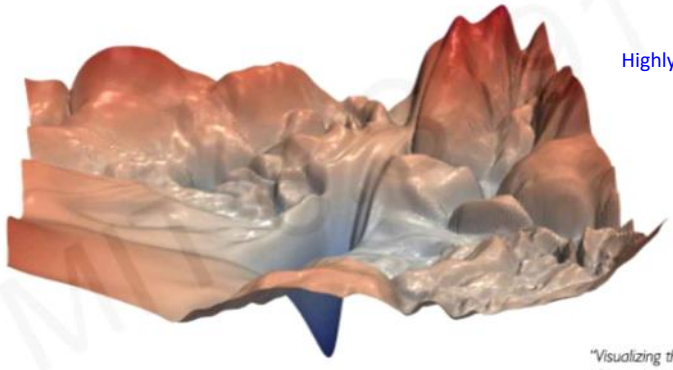
$$\frac{\partial J(W)}{\partial w_1} = \frac{\partial J(W)}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial z_1} * \frac{\partial z_1}{\partial w_1}$$

*Repeat this for **every weight in the network** using gradients from later layers*

eta

Neural Networks in Practice: Optimization

Training Neural Networks is Difficult



Highly non-convex

"Visualizing the loss landscape of neural nets". Dec 2017.

Loss Functions Can Be Difficult to Optimize

Remember:

Optimization through gradient descent

$$\boldsymbol{W} \leftarrow \boldsymbol{W} - \eta \frac{\partial J(\boldsymbol{W})}{\partial \boldsymbol{W}}$$

Loss Functions Can Be Difficult to Optimize

Remember:

Optimization through gradient descent

$$\boldsymbol{W} \leftarrow \boldsymbol{W} - \boxed{\eta} \frac{\partial J(\boldsymbol{W})}{\partial \boldsymbol{W}}$$

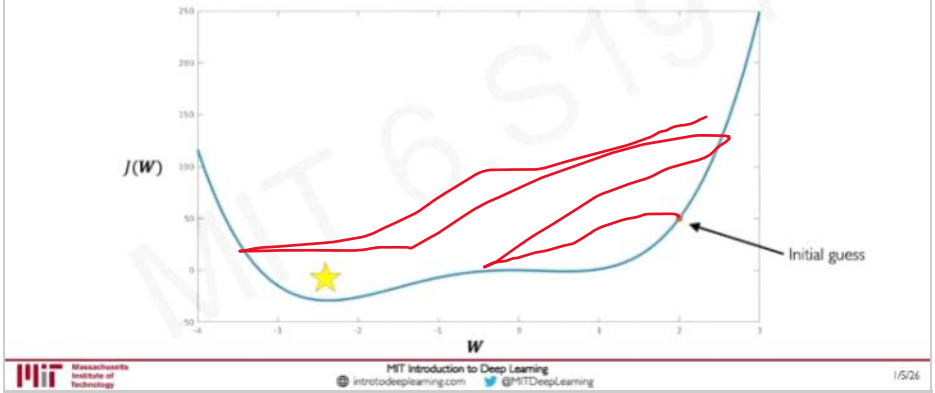
How can we set the learning rate?

Setting the Learning Rate

Small learning rate *converges slowly and gets stuck in false local minima*

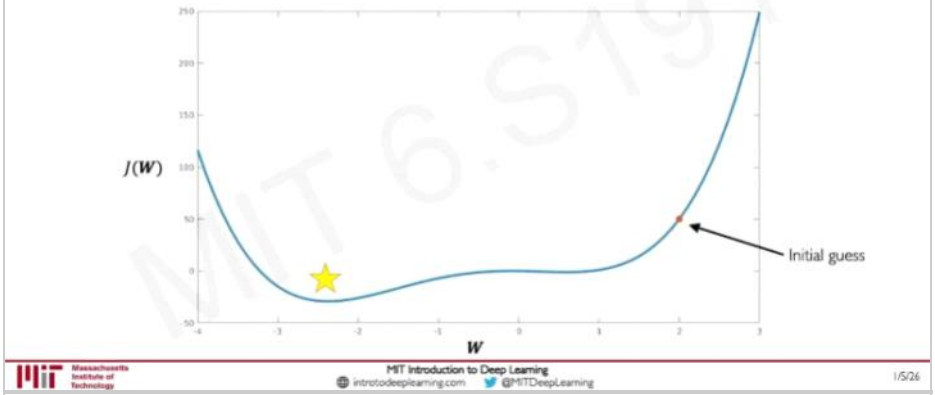
Setting the Learning Rate

Large learning rates *overshoot, become unstable and diverge*



Setting the Learning Rate

Stable learning rates *converge smoothly and avoid local minima*



PROBLEM:
Learning rate has to be large enough to avoid local minima
But small enough to actually reach convergence at the global minimum.

How to deal with this?

Idea 1:

Try lots of different learning rates and see what works "just right"

How to deal with this?

Idea 1:

Try lots of different learning rates and see what works "just right"

Idea 2:

Do something smarter!
Design an adaptive learning rate that "adapts" to the landscape

Adaptive Learning Rates

- Learning rates are no longer fixed
- Can be made larger or smaller depending on:
  - how large gradient is
  - how fast learning is happening
  - size of particular weights
  - etc...

Gradient Descent Algorithms

Algorithm	TF Implementation	Torch Implementation	Reference
SGD	`tf.keras.optimizers.SGD`	`torch.optim.SGD`	Kiefer & Wolfowitz, 1952.
Adam	`tf.keras.optimizers.Adam`	`torch.optim.Adam`	Kingma et al., 2014.
Adadelta	`tf.keras.optimizers.Adadelta`	`torch.optim.Adadelta`	Zeiler et al., 2012.
Adagrad	`tf.keras.optimizers.Adagrad`	`torch.optim.Adagrad`	Duchi et al., 2011.
RMSProp	`tf.keras.optimizers.RMSProp`	`torch.optim.RMSProp`	

Additional details: http://ruder.io/optimizing-gradient-descent/

Putting it all together

```
import tensorflow as tf
model = tf.keras.Sequential([...])
# pick your favorite optimizer
optimizer = tf.keras.optimizer.SGD()
while True: # loop forever
    # forward pass through the network
    prediction = model(x)
    with tf.GradientTape() as tape:
        # compute the loss
        loss = compute_loss(y, prediction)
    # update the weights using the gradient
    grads = tape.gradient(loss, model.trainable_variables)
    optimizer.apply_gradients(zip(grads, model.trainable_variables))
```

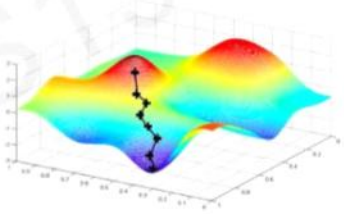
Can replace with any Tensorflow optimizer!

Neural Networks in Practice: Mini-batches

Gradient Descent

Algorithm

1. Initialize weights randomly $\sim\mathcal{N}(0,\sigma^2)$
2. Loop until convergence:
3. Compute gradient, $\frac{\partial J(W)}{\partial W}$
4. Update weights, $W \leftarrow W - \eta \frac{\partial J(W)}{\partial W}$
5. Return weights



Gradient Descent

You need to compute the derivatives of all the weights and for all the datapoints to get this - Very expensive and computationally intensive process

Algorithm

1. Initialize weights randomly $\sim\mathcal{N}(0,\sigma^2)$
2. Loop until convergence:
3. Compute gradient, $\frac{\partial J(W)}{\partial W}$
4. Update weights, $W \leftarrow W - \eta \frac{\partial J(W)}{\partial W}$
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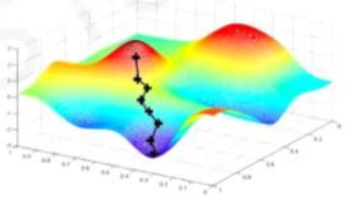


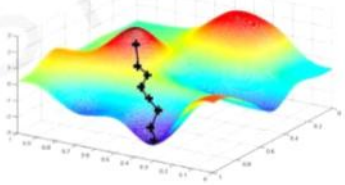
Can be very **computationally intensive** to compute!

Stochastic Gradient Descent

Algorithm

1. Initialize weights randomly $\sim\mathcal{N}(0,\sigma^2)$
2. Loop until convergence:
3. Pick single data point i
4. Compute gradient, $\frac{\partial J_i(W)}{\partial W}$
5. Update weights, $W \leftarrow W - \eta \frac{\partial J(W)}{\partial W}$
6. Return weights

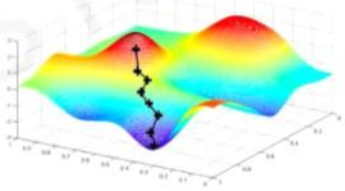
Few data points - common 32, for LLMs batch size may be in millions



Stochastic Gradient Descent

Algorithm

1. Initialize weights randomly $\sim\mathcal{N}(0,\sigma^2)$
2. Loop until convergence:
3. Pick single data point i
4. Compute gradient, $\frac{\partial J_i(W)}{\partial W}$
5. Update weights, $W \leftarrow W - \eta \frac{\partial J(W)}{\partial W}$
6. Return weights



Easy to compute but **very noisy** (stochastic)!

Stochastic Gradient Descent

Algorithm

1. Initialize weights randomly $\sim\mathcal{N}(0,\sigma^2)$
2. Loop until convergence:
3. Pick batch of B data points
4. Compute gradient, $\frac{\partial J(\boldsymbol{W})}{\partial \boldsymbol{W}} = \frac{1}{B}\sum_{k=1}^{B}\frac{\partial J_k(\boldsymbol{W})}{\partial \boldsymbol{W}}$
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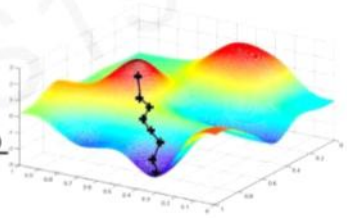

Faster more efficient, but noisy

Stochastic Gradient Descent

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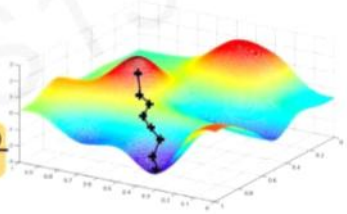


Fast to compute and a much better estimate of the true gradient!

Mini-batches while training

More accurate estimation of gradient

Smoother convergence

Allows for larger learning rates

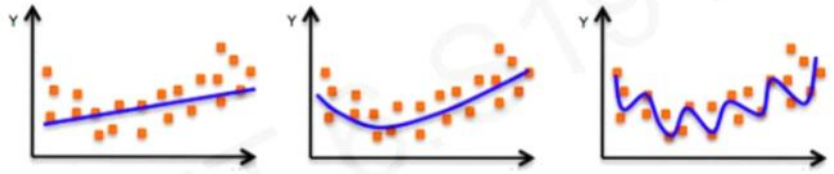
Mini-batches while training

More accurate estimation of gradient
Smoother convergence
Allows for larger learning rates

Mini-batches lead to fast training!
Can parallelize computation + achieve significant speed increases on GPU's

Neural Networks in Practice: Overfitting

The Problem of Overfitting



Underfitting
Model does not have capacity to fully learn the data

Ideal fit

Overfitting
Too complex, extra parameters, does not generalize well

Function not expressive enough to capture the intricacies of data

Regularization

Helps prevent underfitting and overfitting

What is it?

Technique that constrains our optimization problem to discourage complex models

Regularization

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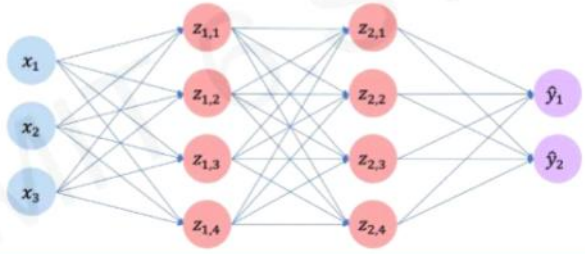
Technique that constrains our optimization problem to discourage complex models

Why do we need it?

Improve generalization of our model on unseen data

Regularization 1: Dropout

- During training, randomly set some activations to 0

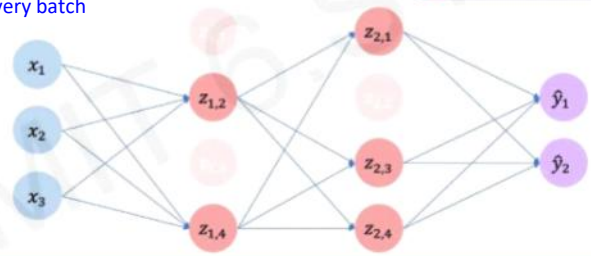


Regularization 1: Dropout

- During training, randomly set some activations to 0
  - Typically 'drop' 50% of activations in layer
  - Forces network to not rely on any 1 node

```
tf.keras.layers.Dropout(p=0.5)
torch.nn.Dropout(p=0.5)
```

Reset on every batch



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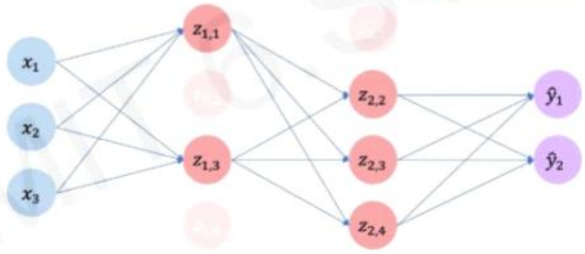

Regularization 2: Early Stopping

- Stop training before we have a chance to overfit

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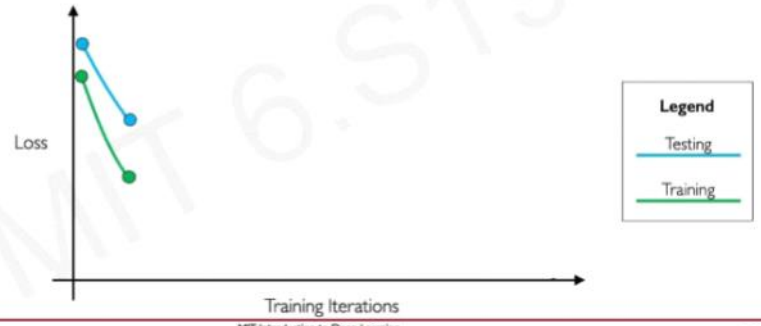

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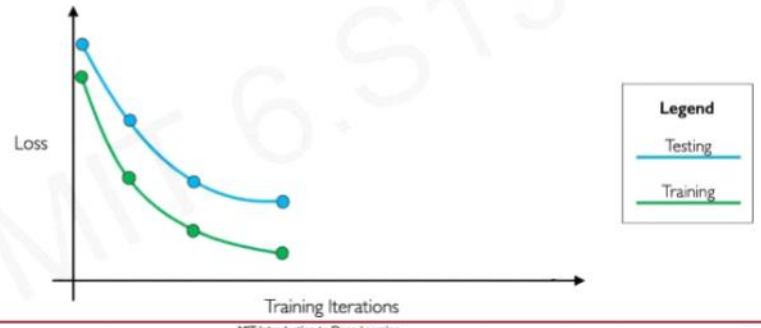

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Core Foundation Review

The Perceptron

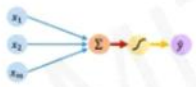
- Structural building blocks
- Nonlinear activation functions

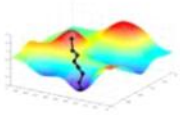
Neural Networks

- Stacking Perceptrons to form neural networks
- Optimization through backpropagation

Training in Practice

- Adaptive learning
- Batching
- Regularization





MIT Introduction to Deep Learning

Lab 1: Introduction to Deep Learning in Python and Music Generation with RNNs

Link to download labs:
http://introtodeeplearning.com#schedule

1. Open the lab in Google Colab
2. Start executing code blocks and filling in the #TODOs
3. Need help? Come to 32-123!